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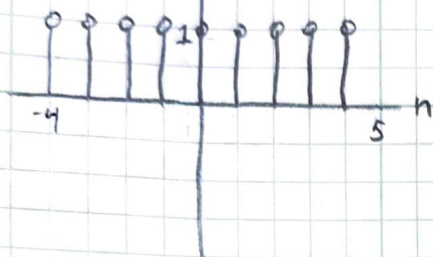
ECEN 380
3/9/2026

PS09: 3.61(b), 3.62(a),
3.68(a)(c), 3.70(a)(d),
MATLAB 3.117

3.61(b)

Use the table of transforms and properties to find the DTFT of the following signal:

$$x[n] = (n-2)(u[n+4] - u[n-5])$$



$$\Rightarrow \begin{cases} 1, & -4 \leq n \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

$$\boxed{1/6}$$

$$x[n] = (n-2) \cdot \begin{cases} 1, & |n| \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

$$C.7: -jn x[n] \xleftrightarrow{\text{DTFT}} \frac{d}{d\omega} X(e^{j\omega})$$

$$(n) x[n] \xleftrightarrow{\text{DTFT}} \frac{d}{d\omega} X(e^{j\omega}) \cdot \frac{1}{-j} = j \frac{d}{d\omega} X(e^{j\omega})$$

$$C.3: x[n] = \begin{cases} 1, & |n| \leq M \\ 0, & \text{otherwise} \end{cases} \xleftrightarrow{\text{DTFT}} X(e^{j\omega}) = \frac{\sin[\omega(\frac{2M+1}{2})]}{\sin(\omega/2)}$$

$$C.7: x[n-n_0] \xleftrightarrow{\text{DTFT}} e^{-j\omega n_0} X(e^{j\omega})$$

$$C.7: ax[n] + by[n] \xleftrightarrow{\text{DTFT}} aX(e^{j\omega}) + bY(e^{j\omega})$$

$$x[n] = n \cdot \begin{cases} 1, & |n| \leq 4 \\ 0, & \text{otherwise} \end{cases} = 2 \cdot \begin{cases} 1, & |n| \leq 4 \\ 0, & \text{otherwise} \end{cases} \cdot \begin{cases} 1, & |n| \leq 4 \\ 0, & \text{otherwise} \end{cases} = 2w[n]$$

$$W(e^{j\omega}) = \frac{\sin(\frac{9}{2}\omega)}{\sin(\omega/2)}$$

$$\text{DTFT}\{nw[n]\} = j \frac{d}{d\omega} \frac{\sin(\frac{9}{2}\omega)}{\sin(\omega/2)}$$

$$\text{DTFT}\{-2w[n]\} = -2 \frac{\sin(\frac{9}{2}\omega)}{\sin(\omega/2)}$$

$$\boxed{\text{DTFT}\{x[n]\} = X(e^{j\omega}) = j \frac{d}{d\omega} \left[\frac{\sin(\frac{9}{2}\omega)}{\sin(\omega/2)} \right] - 2 \cdot \frac{\sin(\frac{9}{2}\omega)}{\sin(\omega/2)}}$$

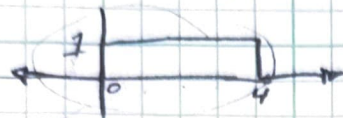
3.62(a)

Use the FT pair:

$$x(t) = \begin{cases} 1, & |t| < 1 \\ 0, & \text{otherwise} \end{cases} \xleftrightarrow{\text{FT}} X(j\omega) = \frac{2\sin(\omega)}{\omega}$$

$$\boxed{2/6}$$

and the FT properties to evaluate the frequency-domain representation of the signal depicted:



$$y(t) = x(t/4), \text{ where } x(t) = \begin{cases} 1, & |t| < 1 \\ 0, & \text{otherwise} \end{cases}$$

$$\text{C.7: } x(at) \xleftrightarrow{\text{FT}} \frac{1}{|a|} X\left(\frac{j\omega}{a}\right)$$

$$X(j\omega) = \frac{2\sin(\omega)}{\omega} \implies \text{FT}\{x(t/4)\} = \frac{1}{|1/4|} \cdot \frac{2\sin(\omega/4)}{\omega/4}$$

$$Y(j\omega) = 4 \cdot \frac{2\sin(4\omega)}{4\omega} = \frac{2\sin(4\omega)}{\omega}$$

$$\boxed{Y(j\omega) = \frac{2\sin(4\omega)}{\omega}}$$

3.68(a)

Determine the frequency and the impulse response for the system described by the following differential equation:

$$(a) \quad \frac{d}{dt} y(t) + 3y(t) = x(t)$$

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class slides:

$$FT \{ h(t) \} = H(j\omega)$$

impulse response

freq. response

;

$$y(t) = h(t) * x(t)$$

$$Y(j\omega) = H(j\omega) X(j\omega)$$

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)}$$

$$C.7: \quad \frac{d}{dt} x(t) \xrightarrow{FT} j\omega X(j\omega)$$

$$(j\omega) Y(j\omega) + 3Y(j\omega) = X(j\omega)$$

$$(j\omega + 3) Y(j\omega) = X(j\omega)$$

$$\frac{Y(j\omega)}{X(j\omega)} = \frac{1}{j\omega + 3} = H(j\omega)$$

$$C.4: \quad x(t) = e^{-at} u(t), \operatorname{Re}\{a\} > 0 \xrightarrow{FT} X(j\omega) = \frac{1}{a + j\omega}$$

$$\operatorname{Inv. FT} \{ H(j\omega) \} \stackrel{\text{inv. FT}}{=} \frac{1}{3 + j\omega} = e^{-3t} u(t) = h(t)$$

$$\text{Impulse response: } h(t) = e^{-3t} u(t)$$

$$\text{Frequency response: } H(j\omega) = \frac{1}{j\omega + 3}$$

3.68(c)

Determine frequency & impulse responses for the system described by the following difference equation:

$$(c) \quad y[n] - \frac{1}{4}y[n-1] - \frac{1}{8}y[n-2] = 3x[n] - \frac{3}{4}x[n-1]$$

$$\text{DTFT}\{h[n]\} = H(e^{j\omega}) ; \quad y[n] = h[n] * x[n]$$

$$Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega})$$

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})}$$

$$\text{C.7: } x[n-n_0] \xrightarrow{\text{DTFT}} e^{-j\omega n_0} X(e^{j\omega})$$

$$Y(e^{j\omega}) - \frac{1}{4}e^{-j\omega}Y(e^{j\omega}) - \frac{1}{8}e^{-j2\omega}Y(e^{j\omega}) = 3X(e^{j\omega}) - \frac{3}{4}e^{-j\omega}X(e^{j\omega})$$

$$Y(e^{j\omega}) \left[1 - \frac{1}{4}e^{-j\omega} - \frac{1}{8}e^{-j2\omega} \right] = X(e^{j\omega}) \left[3 - \frac{3}{4}e^{-j\omega} \right]$$

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{3 - \frac{3}{4}e^{-j\omega}}{1 - \frac{1}{4}e^{-j\omega} - \frac{1}{8}e^{-j2\omega}} = \frac{3 - \frac{3}{4}e^{-j\omega}}{-\frac{1}{8}(e^{-j\omega} + 4)(e^{-j\omega} - 2)}$$

$$\begin{aligned} & \frac{-\frac{1}{8}x^2 - \frac{1}{4}x + 1}{-\frac{1}{8}(x^2 + 2x - 8)} = \frac{-24 + 6e^{-j\omega}}{(e^{-j\omega} + 4)(e^{-j\omega} - 2)} = \frac{A}{(e^{-j\omega} + 4)} + \frac{B}{(e^{-j\omega} - 2)} \\ & \frac{-\frac{1}{8}(x+4)(x-2)}{-\frac{1}{8}(x+4)(x-2)} \quad (x = e^{-j\omega} \text{ for now}) \end{aligned}$$

$$\frac{-24 + 6x}{(x+4)(x-2)} = \frac{A}{(x+4)} + \frac{B}{(x-2)} \Rightarrow -24 + 6x = A(x-2) + B(x+4)$$

$$6x + -24 = Ax - 2A + Bx + 4B$$

$$A + B = 6$$

$$-2A + 4B = -24$$

$$\begin{bmatrix} 1 & 1 & 6 \\ -2 & 4 & -24 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & -2 \end{bmatrix}$$

$$= \frac{8}{(e^{-j\omega} + 4)} + \frac{-2}{(e^{-j\omega} - 2)}$$

$$\text{C.3: } x(e^{j\omega}) = \frac{1}{1 - \alpha e^{j\omega}} \xrightarrow{\text{DTFT}} x[n] \alpha^n u[n], |\alpha| < 1$$

$$H(e^{j\omega}) = \frac{8}{(e^{j\omega} - 4) + (e^{j\omega} - 2)} + \frac{8}{4(1 + \frac{1}{4}e^{j\omega})} = \frac{8}{4(1 - (-\frac{1}{4}e^{j\omega}))} \quad \boxed{5/6}$$

$$= \frac{2}{1 - (-\frac{1}{4}e^{j\omega})}$$

$$\frac{-2}{2(1 - \frac{1}{2}e^{j\omega})} = \frac{1}{1 - \frac{1}{2}e^{j\omega}}$$

$$H(e^{j\omega}) = \frac{2}{1 - (-\frac{1}{4}e^{j\omega})} + \frac{1}{1 - \frac{1}{2}e^{j\omega}}$$

$$h[n] = \text{Inv. DTFT} \left\{ \sqrt{3} \right\} = 2\left(\frac{1}{4}\right)^n u[n] + \left(\frac{1}{2}\right)^n u[n]$$

$$H(e^{j\omega}) = \frac{2}{1 - (-\frac{1}{4}e^{j\omega})} + \frac{1}{1 - \frac{1}{2}e^{j\omega}} \quad (\text{frequency response})$$

$$h[n] = 2\left(\frac{1}{4}\right)^n u[n] + \left(\frac{1}{2}\right)^n u[n] \quad (\text{impulse response})$$

3.70(a) Determine differential-equation description for system w/ frequency response:

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$$H(j\omega) = \frac{2 + 3j\omega - 3(j\omega)^2}{1 + 2j\omega}$$

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{2 + 3j\omega - 3(j\omega)^2}{1 + 2j\omega}$$

$$Y(j\omega)(1 + 2j\omega) = X(j\omega)(2 + 3j\omega - 3(j\omega)^2)$$

$$Y(j\omega) + 2j\omega Y(j\omega) = 2X(j\omega) + 3j\omega X(j\omega) - 3(j\omega)^2 X(j\omega)$$

$$\boxed{y(t) + 2 \frac{d}{dt} y(t) = 2x(t) + 3 \frac{d}{dt} x(t) - 3 \frac{d^2}{dt^2} x(t)}$$

3.70(d) Determine difference-equation description for system w/ frequency response:

$$H(e^{j\omega}) = \frac{1 + e^{-j\omega}}{e^{-j2\omega} + 3}$$

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{1 + e^{-j\omega}}{e^{-j2\omega} + 3}$$

$$Y(e^{j\omega})(e^{-j2\omega} + 3) = X(e^{j\omega})(1 + e^{-j\omega})$$

$$e^{-j2\omega} Y(e^{j\omega}) + 3Y(e^{j\omega}) = X(e^{j\omega}) + e^{-j\omega} X(e^{j\omega})$$

$$\boxed{y[n-2] + 3y[n] = x[n] + x[n-1]}$$

(3.117)

The frequency response of either a continuous or discrete time system described in state-variable form (see Problem 3.87) may be computed with the use of the MATLAB command `freqresp`. The syntax is **`h = freqresp(sys,w)`**, where **`sys`** is the object containing the state-variable description (see Section 2.14) and **`w`** is a vector containing the frequencies at which to evaluate the frequency response. Note that **`freqresp`** applies in general to multiple-input, multiple-output systems, so the output **`h`** is a multidimensional array. For the class of single-input, single-output systems considered in this text and **`N`** frequency points in **`w`**, **`h`** is of dimension 1 by 1 by **`N`**. The command **`squeeze(h)`** converts **`h`** to an **`N`**-dimensional vector that may be displayed with the **`plot`** command. Thus, we may obtain the frequency response by using the following commands:

```
>> h = freqresp(sys,w);  
>> hmag = abs(squeeze(h));  
>> plot(w,hmag)  
>> title('System Magnitude Response');  
>> xlabel('Frequency (rads/sec)'); ylabel('Magnitude');
```

Use MATLAB to plot the magnitude and phase response for the systems with state-variable descriptions given in Problems 3.88 and 3.89.

(Problem 3.88)

(a) $A = \begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix}; b = \begin{bmatrix} 0 \\ 2 \end{bmatrix}; c = [1 \quad 1]; D = [0]$

(b) $A = \begin{bmatrix} 1 & 2 \\ -3 & -4 \end{bmatrix}; b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}; c = [0 \quad 1]; D = [0]$

(Problem 3.89)

(a) $A = \begin{bmatrix} -1/2 & 1 \\ 0 & 1/4 \end{bmatrix}; b = \begin{bmatrix} 0 \\ 1 \end{bmatrix}; c = [1 \quad 0]; D = [1]$

(b) $A = \begin{bmatrix} 1/4 & 3/4 \\ 1/4 & -1/4 \end{bmatrix}; b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}; c = [0 \quad 1]; D = [0]$

```

A = [-2, 0; 0, -1];
b = [0; 2];
c = [1, 1];
D = 0;
sys1 = ss(A,b,c,D,-1);

A = [1, 2; -3, -4];
b = [1; 2];
c = [0, 1];
D = 0;
sys2 = ss(A,b,c,D,-1);

A = [-1/2, 1; 0, 1/4];
b = [0; 1];
c = [1, 0];
D = 1;
sys3 = ss(A,b,c,D,-1);

A = [1/4, 3/4; 1/4, -1/4];
b = [1; 1];
c = [0, 1];
D = 0;
sys4 = ss(A,b,c,D,-1);

% w = linspace(0, 100, 500);
% w = logspace(1,5,500);
w = logspace(-1,2,500);

subplot(2, 1, 1);
h = freqresp(sys1,w);
hmag = abs(squeeze(h));
semilogx(w,hmag)
title('System Magnitude Response for Problem 3.88(a)');
xlabel('Frequency (rads/sec)'); ylabel('Magnitude');
grid on;
subplot(2, 1, 2);
hphase = angle(squeeze(h)) * 180/pi; % convert to degrees
semilogx(w,hphase)
title('System Phase Response for Problem 3.88(a)')
xlabel('Frequency (rads/sec)'); ylabel('Phase (deg)')

```



```

figure
subplot(2, 1, 1);
h = freqresp(sys2,w);
hmag = abs(squeeze(h));
semilogx(w,hmag)
title('System Magnitude Response for Problem 3.88(b)');
xlabel('Frequency (rads/sec)'); ylabel('Magnitude');
grid on;
subplot(2, 1, 2);
hphase = angle(squeeze(h)) * 180/pi; % convert to degrees
semilogx(w,hphase)
title('System Phase Response for Problem 3.88(b)')
xlabel('Frequency (rads/sec)'); ylabel('Phase (deg)')
grid on;

```

```

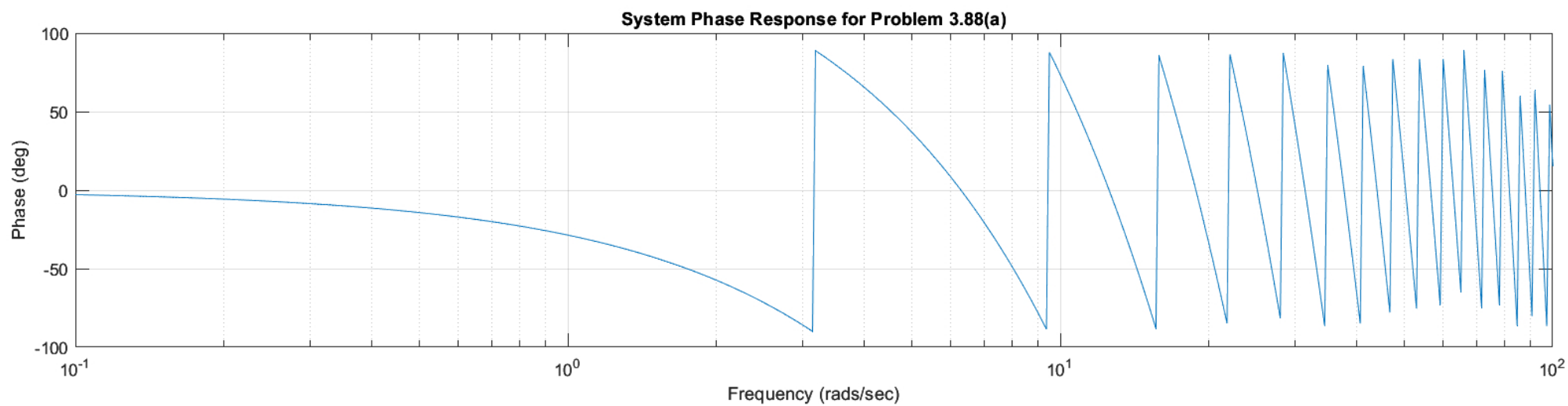
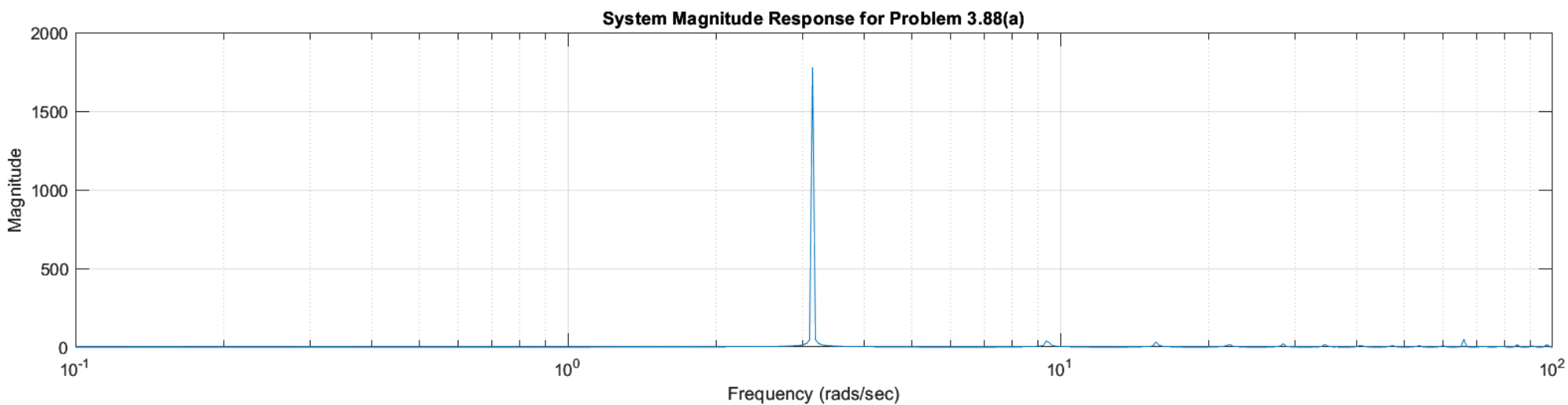
figure
subplot(2, 1, 1);
h = freqresp(sys3,w);
hmag = abs(squeeze(h));
semilogx(w,hmag)
title('System Magnitude Response for Problem 3.89(a)');
xlabel('Frequency (rads/sec)'); ylabel('Magnitude');
grid on;
subplot(2, 1, 2);
hphase = angle(squeeze(h)) * 180/pi; % convert to degrees
semilogx(w,hphase)
title('System Phase Response for Problem 3.89(a)')
xlabel('Frequency (rads/sec)'); ylabel('Phase (deg)')
grid on;

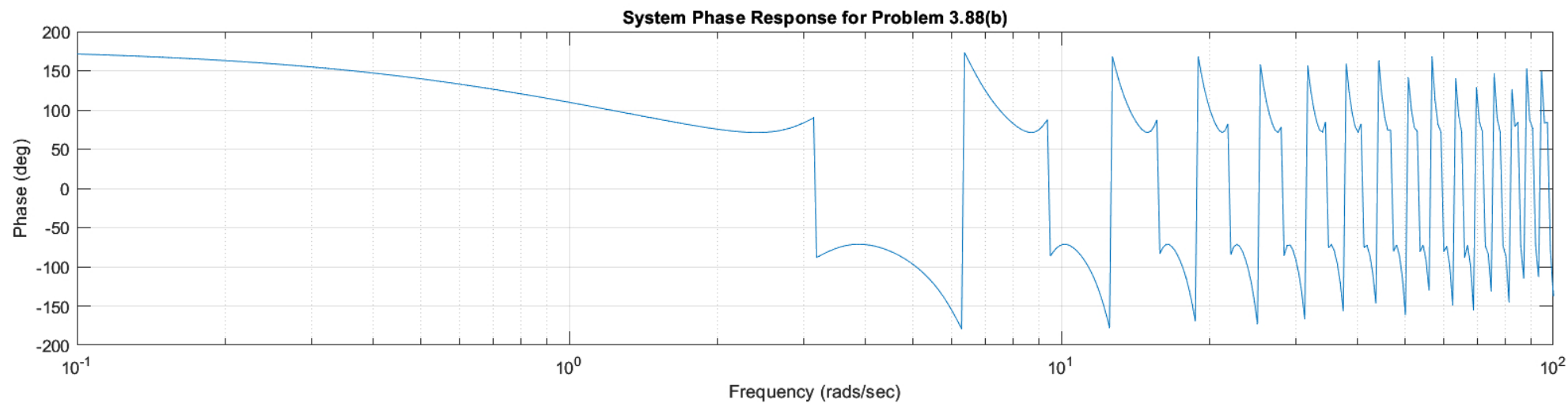
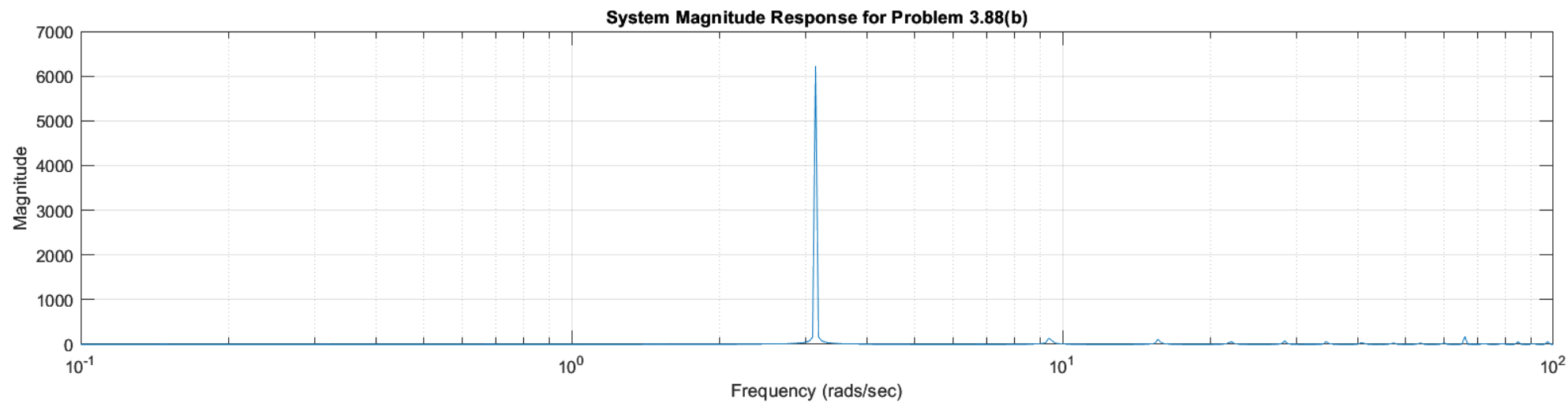
```

```

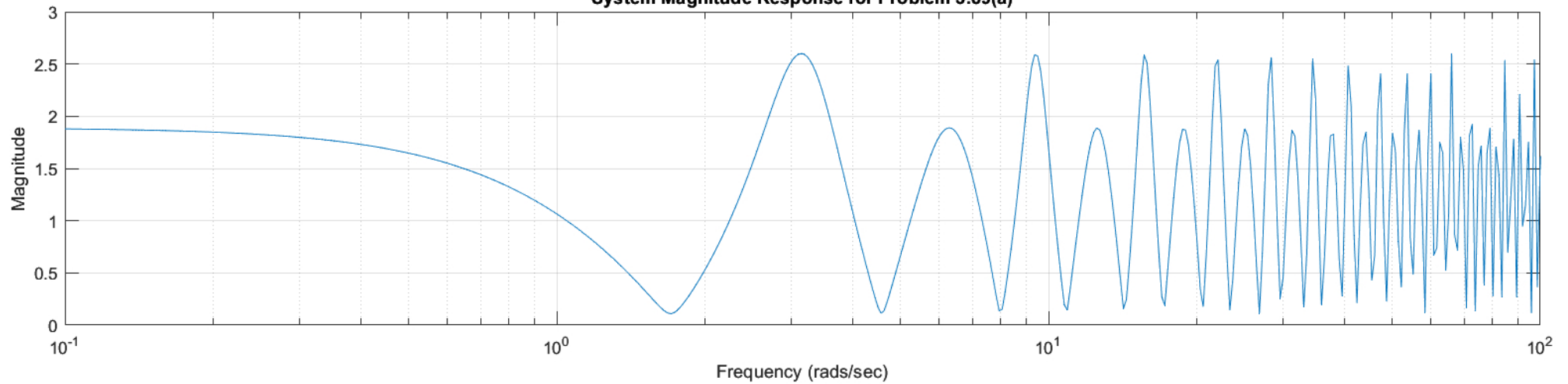
figure
subplot(2, 1, 1);
h = freqresp(sys4,w);
hmag = abs(squeeze(h));
semilogx(w,hmag)
title('System Magnitude Response for Problem 3.89(b)');
xlabel('Frequency (rads/sec)'); ylabel('Magnitude');
grid on;
subplot(2, 1, 2);
hphase = angle(squeeze(h)) * 180/pi; % convert to degrees
semilogx(w,hphase)

```

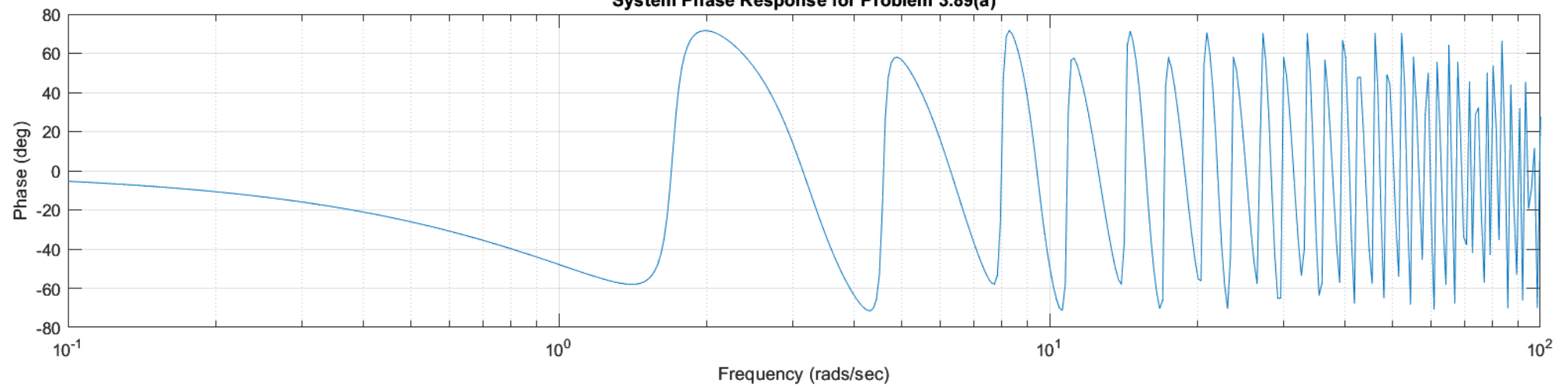




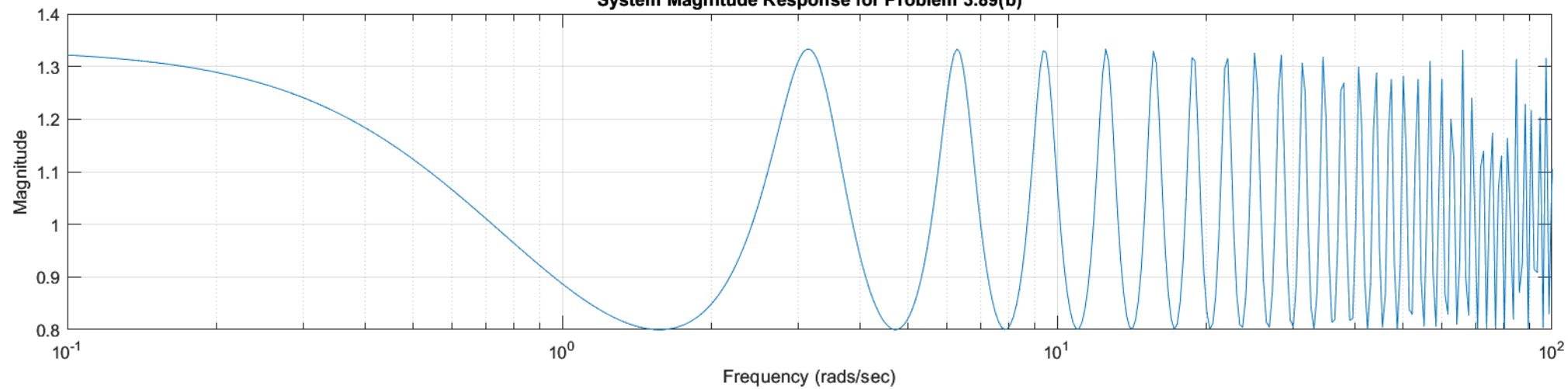
System Magnitude Response for Problem 3.89(a)



System Phase Response for Problem 3.89(a)



System Magnitude Response for Problem 3.89(b)



System Phase Response for Problem 3.89(b)

